

GLOBAL MULTIROW COERCIVITY CERTIFICATE

Exact finite-state Schur certification, dual optimality,
and arbitrary-height coercivity

**EXACT GLOBAL MULTIROW COERCIVITY CERTIFICATE
UNIQUE TARGET, NONUNIQUE AUXILIARY COMPLETIONS**

Release version 1.0

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Clean-room status: the complete mathematical certificate was independently regenerated after repair of the exact source-functional provenance packet.

Verification package SHA-256

8fb766a63858d502c19b9dc2121df76d
b249d05af417fc0106a70371233e3d9a

Verifier source SHA-256

6b5b4a83ceebe49f1843c5de5bb99983
5855a631d80a1c0b98e6a8f3a318586d

Exact source-functional packet SHA-256

bdced7e61685cd78730cf94cf325f597
a48fa76ad6b244c3b1efc00e6cb18fdb

Scope is limited to the fixed interval, shifts, transition, source operator, and target specified in this certificate. No digital signature is claimed; integrity is represented by SHA-256 hashes.

This PDF intentionally does not contain its own final SHA-256. Final publication hashes are external.

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1. Executive Certification Statement

The certificate records a fully reproduced exact semidefinite proof for the fixed multirow operator instance. The clean-room verifier returned:

A. CLEAN-ROOM GLOBAL CERTIFICATE FULLY REPRODUCED AFTER SOURCE REPAIR

The target is unique and the auxiliary Schur completions are nonunique.

The optimal target face has dimension zero. The auxiliary completion variables remain locally nonunique with dimensions 91, 105, and 225, totaling 421.

The replay regenerated the isolated root, all sixteen finite-state matrices, all 44 scalar inequalities, active dual factors, stationarity, complementary slackness, primal-dual equality, target rank, Schur induction, and support geometry.

2. Formal Theorem

Let

$$H_2 = 1260233493541651302622586638887192915605138797326.$$

For the fixed target shifts

$$(1,0), (3,0), (0,1), (-1,1), (-3,1),$$

and the uniquely certified target c^* , define

$$W_2(c^*) = I + \sum_{j=1}^5 c_j^* D_{s_{2,j}}.$$

Then, in Löwner order,

$$\frac{1}{2} I \leq W_2(c^*) \leq 2 I \quad \text{on } [0, H_2].$$

The exact transition and source operator are

$$3^{96}/4^{76} \rightarrow 3^{101}/4^{80},$$

$$W_1 = I - (127/768)D_c.$$

No assertion is made for other intervals, target shifts, coefficient vectors, source operators, or transition architectures.

3. Problem Definition

The exact arithmetic parameters are:

Quantity	Exact value
u	84630924865131074905698556580580798450390023027
v	1207608862735509693486589162974540624304762473895
H2	1260233493541651302622586638887192915605138797326
$\delta = v - 14u$	22775914623674644806809370846409445999302151517
$R_H = H_2 - 14u$	75400545429816253942806846759061737299678474948

The verified inequalities are $0 < \delta < R_H < u$, $14u \leq H_2 < 15u$, and $\gcd(u,v)=1$. The exact automaton is

$$T = \{(15,0,15), (15,0,14), (15,1,15), (14,1,15)\}.$$

For $n \in \{14,15\}$,

$$\begin{aligned} T_L^{(n)}(c) &= (\tfrac{1}{2} + 2 \Sigma c_j) I_n - c_1 A_1^{(n)} - c_2 A_3^{(n)}. \\ T_U^{(n)}(c) &= (1 - 2 \Sigma c_j) I_n + c_1 A_1^{(n)} + c_2 A_3^{(n)}. \end{aligned}$$

Transition blocks use the exact $J_d^{n,m}$ shift matrices with offsets $14+\varepsilon$, $13+\varepsilon$, and $11+\varepsilon$.

4. Exact Source Functional

The authoritative problem-definition packet supplies the following reduced rational values independently of the proof package.

Symbol	Numerator	Denominator
S	1476402264857835952197482207371347506998319083517	384
G0	3928012289356872196325154005257155416462979638113	131072
E1	464640253293273745091699662807230736035856379673	32768
E2	101806105062068953205793323512757807017352137759	4096
E3	3871311037086588629779609348556262785118720864965	65536
E4	7485556487300223380862036258672975760839595190779	131072
E5	6851895575569244989411973667925987545523611892851	131072

$$\begin{aligned} E &= (E_1, E_2, E_3, E_4, E_5), \quad \text{minimize } E \cdot c, \\ \rho(c) &= (G_0 + E \cdot c)/S. \end{aligned}$$

Every fraction is reduced and S is positive. All seven values matched the candidate proof package after the source packet had been independently sealed.

5. Isolated Target

The target is the unique real root in

$$B = \Xi + [-10^{-100}, 10^{-100}]^{37}.$$

The five target coordinates are displayed to 24 decimal places; the source packet contains the full center.

Coordinate	Displayed center
c1	0.227533338966018337888847
c2	-0.019079301528073732718368
c3	-0.373639038703067559864387
c4	0.116305780468658569317658
c5	0.007255491697890803218255

Variable order: $c_1, \dots, c_5$; $p_{0,0}, \dots, p_{0,14}$; $p_{1,0}, \dots, p_{1,14}$; τ ; lam . The center TSV uses “lambda” at index 36. The verified alias $\text{lambda} \rightarrow \text{lam}$ is lexical only, with no permutation or transformation of Ξ , A, F, or JF.

The Krawczyk inclusion and an independent inverse/contraction check both proved existence and uniqueness in B.

6. Active Lower-15 Completion

Let $U=[p_0 \ p_1]$, $G=U^T U$, and $\Pi=I-UG^{-1}U^T$. The active completion is

$$K_{L,15} = K_0 + \Pi S_{\text{comp}} \Pi.$$

S_{comp} is the rational completion matrix and is not the source normalization S .

The exact kernel-action identities and both interval LDL^T and interval Cholesky certificates were regenerated.

Matrix	Status	Certified lower bound / positive gap
$K_{L,15}$	PD	4.04473726348265527E-8
$T_L^{(15)} - K_{L,15}$	PD	4.66337066816059830E-7
$M_{L,15,0,15}$	PSD, nullity 1	1.50407570574840730E-9
$M_{L,15,1,15}$	PSD, nullity 1	1.16203550950293358E-8

7. Full Finite-State Primal Feasibility

All sixteen state and transition matrices were regenerated. Every intended positive-definite matrix passed interval LDL^T and interval Cholesky. The only singular matrices are the two active lower self-transitions, each with nullity one.

Matrix	Dim.	Status	Null.	Certified bound
K L14	14	PD	0	9.8238066927132E-5
T L14 minus K L14	14	PD	0	3.7273646981640E-4
K L15	15	PD	0	4.0447372634827E-8
T L15 minus K L15	15	PD	0	4.6633706681606E-7
M L 15 0 15	30	PSD	1	1.5040757057484E-9
M L 15 0 14	29	PD	0	1.0005435444295E-8
M L 15 1 15	30	PSD	1	1.1620355095029E-8
M L 14 1 15	29	PD	0	1.7169086421608E-8
K U14	14	PD	0	3.5267979896268E-2
T U14 minus K U14	14	PD	0	3.5267979896268E-2
K U15	15	PD	0	3.2908356435185E-2
T U15 minus K U15	15	PD	0	3.2908356435185E-2
M U 15 0 15	30	PD	0	3.7289784170619E-3
M U 15 0 14	29	PD	0	1.6111538548937E-2
M U 15 1 15	30	PD	0	1.5573537394693E-2
M U 14 1 15	29	PD	0	1.6111538548937E-2

Appendix A reproduces the full decimal lower-bound strings.

8. Finite Scalar Constraints

All 22 witnesses contribute two strict inequalities, for 44 strict constraints. Both minima are attained at witness v3.

minimum lower slack > 0.08622700619957055163924767398647661736708233926517

minimum upper slack > 2.9137729938004294483607523260135233826329176607348

The source packet selects w3 with $N=2470$ and $D=(276,960,5036,4920,4472)$. The alternate inherited row was not silently substituted. The complete audit is in `regenerated_finite_slacks_v2.tsv`.

9. Dual Certificate

$$Y_0 = \alpha z_0 z_0^T, \quad Y_1 = \beta z_1 z_1^T,$$

$$\beta = E_1/\Phi_1, \quad \alpha = \lambda^2\beta.$$

Verified positive interval for α :

[23949798656841640958393682538830470103282117.61891183362591497206023498881068168919582293497977192532526423706468631979207286480298,
23949798656841640958393682538830470103282117.61891183362591497206023498881068168919582293497977192558092253492141678463826337774738]

Verified positive interval for β :

[25241191788194110649893730064173021174157108.33966244422994530350147417745491751483201014527937947968384270703371121826979090368855,
25241191788194110649893730064173021174157108.33966244422994530350147417745491751483201014527937947994292120540301129476112064554419]

Thus Y_0 and Y_1 are rank-one positive semidefinite matrices. All inactive matrix duals and finite multipliers are zero.

10. Stationarity and Complementary Slackness

With $D=\text{diag}(\alpha,\beta)$, $Q=UR$, and $RDR^T=D$, the exact K-stationarity identity is

$$\alpha(p_0p_0^T - q_0q_0^T) + \beta(p_1p_1^T - q_1q_1^T) = 0.$$

For $j=1,\dots,5$, exact c-stationarity reduces to $E_j=\beta\Phi_j$; for $j=2,\dots,5$ the denominator-cleared relations are

$$E_j\Phi_1 - E_1\Phi_j = 0.$$

Exact kernel equations imply

$$M_0Y_0=0, \quad M_1Y_1=0, \quad \langle M_0, Y_0 \rangle = \langle M_1, Y_1 \rangle = 0.$$

11. Primal-Dual Equality and Objective

The generated Lagrangian uses $L=E \cdot c - \Sigma(Y_s, F_s) - \Sigma \mu_r g_r$. Exact stationarity cancels all variable terms. The remaining constant is

$$C_{\text{dual}} = \frac{1}{2}(\alpha\|q_0\|^2 + \beta\|q_1\|^2),$$

$$E \cdot c^* = -C_{\text{dual}}.$$

Verified normalized objective interval:

$\rho_* \in ($
0.0045959787401878324127301284818766727889572120896095090758308523957239403075589880157367981561252398505898223273517246222,
0.00459597874018783241273012848187667278895721208960950907583085239572394030755898801573679815612523989261333858994889408665)

$$\rho_* - \rho_{\text{3row}} > 1.1742526364832112540441734819190401967290896095 \times 10^{-6}.$$

12. Target Uniqueness and Auxiliary Nonuniqueness

For any competing optimal point, rank-one complementarity gives the affine kernel system. Exact rank verification yields

$$\text{rank}(A_K) = 29, \quad \text{rank}([A_K \ A_c]) = 34.$$

Therefore the five target columns add five independent directions modulo the state-matrix columns.

$$c' = c^*, \quad \dim(\text{optimal target face})=0.$$

Auxiliary completions are not unique: the local dimensions are 91 for $K_{L,15}$, 105 for $K_{L,14}$, and 225 for the upper pair, totaling 421.

13. Arbitrary-Height Schur Induction

For $\sigma \in \{L, U\}$, assume

$$S_k \geq K_{\sigma, n_k} > 0.$$

Then inversion reverses order:

$$S_k^{-1} \leq K_{\sigma, n_k}^{-1}.$$

For the next transition block C_k ,

$$\begin{aligned} S_{k+1} &= T_{\sigma}^{(n_{k+1})} - C_k^T S_k^{-1} C_k \\ &\geq T_{\sigma}^{(n_{k+1})} - C_k^T K_{\sigma, n_k}^{-1} C_k \geq K_{\sigma, n_{k+1}} > 0. \end{aligned}$$

The final step is the Schur complement of the certified transition LMI. Singular active transition blocks cause no obstruction because $K_{L,15}$ is strictly positive definite.

14. Support-Geometry Closure

The vertical components of the five shifts are 0 or 1, so the maximum vertical displacement is one. No edge crosses a missing row. Connected occupied-row sequences correspond to automaton paths. Boundary-trimmed rows are principal submatrices, and arbitrary finite supports decompose into direct sums of connected components. Principal submatrices and finite direct sums preserve positive semidefiniteness.

Combined with the lower and upper Schur inductions, this proves the operator inequalities for every finite support on $[0, H_2]$.

15. Final Theorem and Limitations

$\frac{1}{2} I \leq W_2(c^*) \leq 2 I \quad \text{on } [0, H_2].$

Certified scope: $H_2=1260233493541651302622586638887192915605138797326$; shifts (1,0), (3,0), (0,1), (-1,1), (-3,1); transition $3^{96}/4^{76} \rightarrow 3^{101}/4^{80}$; source operator $W_1 = I - (127/768)D_c$.

No assertion is made for other intervals, shifts, coefficient vectors, source operators, transition architectures, or residual problems. This release is hash-sealed but not digitally signed.

16. Reproducibility Statement

Run the verifier from a directory containing the source packets:

```
python verify_global_certificate_v2.py \
--source-primal primal_schur_branch_packet.zip \
--source-root primal_anchored_rotation_root_isolation.zip \
--source-active active_lower15_schur_completion_packet.zip \
--source-full-primal full_finite_state_primal_feasibility_packet.zip \
--source-functional exact_source_functional_source_packet.zip \
--candidate-final final_global_multirow_coercivity_proof_packet.zip \
--output clean_room_verification_v2
```

Environment: Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0. Successful verification must reproduce:

A. CLEAN-ROOM GLOBAL CERTIFICATE FULLY REPRODUCED AFTER SOURCE REPAIR

17. Provenance and Repair History

1. The first clean-room audit stopped because exact S and G_0 were absent from the first four source packets. No downstream theorem stage was executed.

2. A separate authoritative source-functional packet was created from fixed problem-instance data. Candidate-final data were not used as the source.
3. After sealing the source packet, S , G_0 , and $E_1, \dots, E_5$ were compared against the proof package and matched exactly.
4. The complete clean-room replay then returned Decision A.
5. The λ/lam difference was a lexical alias at index 36, not a permutation or mathematical discrepancy.

Appendix A. Exact Certified State Bounds

For the two PSD active transitions, the listed value is the certified positive spectral gap from the strengthened-matrix argument.

Matrix	Exact decimal lower bound
K L14	0.00009823806692713221382962292777943574095702274928510635214883997436204007399258781721865133433564991878
T L14 minus K L14	0.0003727364698164028507527419252500203465565390097517626846599108855432090228269157325425917228276606548
K L15	0.0000000404473726348265526799734079028055370172689180847611467381138232128403004534644152218910372851081637
T L15 minus K L15	0.000000466337066816059829677438733099298878884172763179763481587885543396614047605154454105647041498166115
M L 15 0 15	0.000000001504075705748407297531317421148675937941847784682166158084174190107664081260824319889910528939131845
M L 15 0 14	0.00000001000543544429477199876573565137013922032501684933879195317012086077578616558266920983973309426061236
M L 15 1 15	0.00000001162035509502933576286924045947857457519872019447962046017730428740219234393644432157248074903509518
M L 14 1 15	0.00000001716908642160788529458915052911692709851791891601052616009140864737013577260673449124173670011803302
K U14	0.0352679798962675052870786526695110361729758949400380220902154222354506323913183542596955447133027872
T U14 minus K U14	0.03526797989626750528707865266951103617297589494003802209021542223545063239131835425969554471330278711
K U15	0.03290835643518461274101801343118584945000100714880122395936784463462182690719716049292385601170504935
T U15 minus K U15	0.03290835643518461274101801343118584945000100714880122395936784463462182690719716049292385601170504926
M U 15 0 15	0.003728978417061881020349611121596389158351653533543296733124760154650476152211047677014006247629437109
M U 15 0 14	0.0161115385489372691948811236438685619448774265810636843650961328578950013535271297410965891894260535
M U 15 1 15	0.01557353739469348987446415262964128968370734059642585843492293771247416670052637362377170543268165509
M U 14 1 15	0.01611153854893726473795138032813005019564040450920003178072801174468554043538878743986514017204064289

Appendix B. Authoritative Input Hashes

The final PDF hash and complete release-member hashes are external in MANIFEST.json, SHA256SUMS.txt, and the release ZIP sidecar. This avoids cryptographic self-reference.

Artifact	SHA-256
primal_schur_branch_packet.zip	50817a80ec3e13832e682e2215c2d511458542101d21ce8474eac0e6ba6489a1
primal_anchored_rotation_root_isolation.zip	40164cc6d65601053ae30dfd63fad7bc6cd662141d4e28b2ebd47774571f0fc7
active_lower15_schur_completion_packet.zip	1f7abc5024948777850335f7cb422c58f60c9efa64e9763038d4454b5ee1c16c
full_finite_state_primal_feasibility_packet.zip	86475b452e86e93047937cd267551bffa7a08c80c87f8df64623bbc2ea14974
exact_source_functional_source_packet.zip	bdced7e61685cd78730cf94cf325f597a48fa76ad6b244c3b1efc00e6cb18fdb
final_global_multirow_coercivity_proof_packet.zip	cce1bde9bb54f2bcd1e7d2be61c91256ce9cbd8c529ae1000e1dd2d948df2e87
clean_room_global_certificate_verification_v2.zip	8fb766a63858d502c19b9dc2121df76db249d05af417fc0106a70371233e3d9a
verify_global_certificate_v2.py	6b5b4a83ceeba49f1843c5de5bb999835855a631d80a1c0b98e6a8f3a318586d

Certificate decision

FORMAL GLOBAL COERCIVITY CERTIFICATE - UNIQUE TARGET, NONUNIQUE AUXILIARY COMPLETIONS

End of certificate.